

NP-Complete Reduction Proof

Problem

$$3COLOR = \{\langle G \rangle \mid G \text{ is 3-colorable}\}$$

We prove that $3COLOR$ is NP-complete by reducing from $3SAT$.

Step 1: Show that $3COLOR \in NP$

A certificate for $3COLOR$ is a proposed coloring of the vertices of G using three colors.

For example, the certificate assigns each vertex one of the colors:

$$\text{Colors} = \{\text{Red, Yellow, Blue}\}$$

To verify the certificate, we check every edge (u, v) in the graph and make sure that u and v have different colors. If every edge connects vertices of different colors, then the coloring is valid. If any edge has endpoints with the same color, then the coloring is invalid.

If G has n vertices and m edges, checking the coloring takes $O(n)$ time to check that each vertex has a valid color and $O(m)$ time to check all edges. Therefore, the total verification time is:

$$O(n + m)$$

which is polynomial. Thus,

$$3COLOR \in NP$$

Step 2: Choose a known NP-complete problem

We reduce from $3SAT$, which is known to be NP-complete.

$3SAT$ is the problem of deciding whether a Boolean formula, where each clause has three literals, can be made true by some assignment of True/False values.

So we will show:

$$3SAT \leq_m 3COLOR$$

Step 3: Show the reduction

Let ϕ be an instance of $3SAT$. We construct a graph G such that:

$$\phi \text{ is satisfiable} \iff G \text{ is 3-colorable}$$

The graph G is built using three types of gadgets: a palette gadget, variable gadgets, and clause gadgets.

Palette gadget

First, create a triangle with three special vertices:

$$T, F, B$$

Since these three vertices form a triangle, they must all receive different colors in any valid 3-coloring. We interpret these colors as:

$$T = \text{True}, \quad F = \text{False}, \quad B = \text{Base}$$

Variable gadget

For each variable x_i in ϕ , create two vertices:

$$x_i \quad \text{and} \quad \overline{x_i}$$

Connect x_i to $\overline{x_i}$. Also connect both x_i and $\overline{x_i}$ to B .

Because both x_i and $\overline{x_i}$ are connected to B , neither can use the Base color. Therefore, they must use either the True color or the False color.

Since x_i and $\overline{x_i}$ are connected to each other, they cannot have the same color. Thus, one must be True and the other must be False. This correctly represents a Boolean variable and its negation.

Clause gadget

For each clause in ϕ , such as:

$$(\ell_1 \vee \ell_2 \vee \ell_3)$$

we add a clause gadget connected to the three literal vertices:

$$\ell_1, \ell_2, \ell_3$$

The clause gadget is constructed so that it is 3-colorable exactly when at least one of its three input literals is colored True.

This matches the behavior of a 3SAT clause, since a clause is satisfied exactly when at least one of its literals is true.

Polynomial-time construction

The construction can be computed in polynomial time.

For each variable, we add a constant number of vertices and edges. For each clause, we add one clause gadget, which also has constant size. Therefore, if ϕ has n variables and m clauses, the graph G has size polynomial in $n + m$.

Thus, the transformation from ϕ to G is polynomial-time computable.

Correctness of the reduction

We now prove:

$$\phi \text{ is satisfiable} \iff G \text{ is 3-colorable}$$

Forward direction

Assume ϕ is satisfiable.

Then there exists a truth assignment that makes every clause of ϕ true.

Use this truth assignment to color the variable vertices of G . If x_i is true, color x_i with the True color and $\overline{x_i}$ with the False color. If x_i is false, color x_i with the False color and $\overline{x_i}$ with the True color.

Since every clause in ϕ is satisfied, each clause has at least one true literal. Therefore, each clause gadget has at least one input colored True, so each clause gadget can be properly 3-colored.

Thus, the entire graph G is 3-colorable.

Therefore,

$$\phi \text{ is satisfiable} \implies G \text{ is 3-colorable}$$

Reverse direction

Assume G is 3-colorable.

The palette triangle forces the vertices T , F , and B to all have different colors. Therefore, these colors represent True, False, and Base.

For each variable x_i , the vertices x_i and $\bar{x}_i$ are both connected to B , so neither can be colored Base. They must be colored either True or False.

Since x_i and $\bar{x}_i$ are connected to each other, they must have opposite colors. Therefore, the coloring gives a valid truth assignment for the variables of ϕ .

Each clause gadget is 3-colorable only when at least one of its input literals is colored True. Since the whole graph is 3-colorable, every clause gadget must have at least one true literal.

Therefore, every clause of ϕ is satisfied.

Thus,

$$G \text{ is 3-colorable} \implies \phi \text{ is satisfiable}$$

Step 4: Conclusion

We have shown that ϕ can be transformed into a graph G in polynomial time, and that:

$$\phi \text{ is satisfiable} \iff G \text{ is 3-colorable}$$

Therefore,

$$3SAT \leq_m 3COLOR$$

Since $3SAT$ is NP-complete, this shows that $3COLOR$ is NP-hard.

We also showed that:

$$3COLOR \in NP$$

Therefore,

$3COLOR \text{ is NP-complete}$